

/\*

## -----French-----

### Exercice3

- 1). Ecrire une fonction récursive *Premier* permettant d'afficher tous les nombres premiers qui sont inférieur au nombre entier passé en paramètre,
- 2). Allouer dynamiquement la mémoire pour le tableau renvoyant le résultat.
- 3). Ecrire un programme principal de test.

## -----English-----

### Exercise3

- 1). Write a recursive function *Prime* to display all prime numbers which are less than the whole number passed as a parameter,
- 2). Dynamically allocate memory for array returning the result.
- 3). Write a main test program.

\*/

```
#include <stdio.h>
#include <stdlib.h>

int *allocate(int *prime, int length);
int primef(int i, int iByTwo); // iByTwo = i / 2
void find_primes(int number);

int main()
{
    int number;
    printf("NUmber = ");
    scanf("%d", &number);
    find_primes(number);
    return 0;
}

int *allocate(int *prime, int length)
{
    prime = (int *)malloc(length * sizeof(int));
    return prime;
}

int primef(int i, int iByTwo)
{
    if (iByTwo == 1)
        return 1;
    if (i % iByTwo == 0)
        return 0;
    else
        return primef(i, iByTwo - 1);
}

void find_primes(int number)
{
    int *prime, length, i, j;
    prime = allocate(prime, length);
    for (i = 2; i <= number; i++)
    {
```

```
if (primef(i, i / 2) == 1)
{
    prime[j] = i;
    length++;
    printf("%d\t", *(prime + j));
    length++;
    j++;
}
}
```

/\*

**Edit by:** MOHAMMED ELIDRISSI LAOUKILI  
#idriCoding | #idriLogic | #idri  
2021/2022

لا اله الا الله

\*/